

Assignment 1

Topic: Linux System Monitoring Tool Using the `/proc` Filesystem

Autumn 2026-2027

Submission Deadline: 14th August EOD

Course: Advances in OS Design

Total Marks: 100

The `/proc` filesystem is a virtual filesystem provided by the Linux kernel. It does not exist on disk. Instead, the kernel generates its contents dynamically whenever you access it. In this assignment, you will build a command-line utility that reads directly from these files to report CPU, memory, and process information, without relying on external utilities such as `ps`, `top`, or `free`. Your code **should not** make use of Linux Performance Monitoring Tools.

1. CPU and Memory Information

Read data from `/proc/cpuinfo`, `/proc/meminfo`.

Output

- CPU model and number of logical processors
- Total, Free, and Available memory
- Percentage of memory currently in use

2. CPU Utilization and System Statistics

Read data from `/proc/stat`, `/proc/uptime`, `/proc/loadavg`.

Output

- CPU utilization, computed over a 1-second sampling interval
- System uptime
- 1-, 5-, and 15-minute load averages

3. Process Information

Read data from the numeric directories under `/proc`, `/proc/[pid]/status`.

Output

For every running process, display:

- PID
- Process name
- Parent PID (PPID)
- Process state
- Number of threads
- Virtual memory size (`VmSize`)

4. Top Resource-Consuming Processes

Read data from `/proc/[pid]/status`, `/proc/[pid]/stat`.

Output

- Top 5 processes by memory usage, using `VmRSS`
- Top 5 processes by CPU usage, computed from values in `/proc/[pid]/stat` sampled over a 1-second interval

5. Live Mode

Read data from the same `/proc` files used in the previous tasks. In this mode, the screen must refresh once every **T seconds**, similar to `top`, and must continue running until the user presses `Ctrl+C`. On every refresh, the display must update:

- CPU usage
- Memory usage
- Process list
- Top 5 CPU-consuming processes
- Top 5 memory-consuming processes

Mode	Command
Live mode	<code>./proc_monitor -T <interval></code>
Static mode	<code>./proc_monitor</code>

Submission Requirements

Submission must be on the CSE Moodle Platform. In this assignment you are free to choose any programming language of your choice. Only one member of each team should submit a ZIP file containing the following:

- Source code with comments
- A Makefile (or equivalent build script) for compilation
- A README file with clear instructions to compile and run the tool in both modes as well as team member names and Roll Numbers

Evaluation

Component	Marks
CPU & Memory Information	15
CPU Utilization & System Statistics	15
Process Information	15
Top Resource-Consuming Processes	20
Live Mode	30
Documentation + README	5

Sample Expected Output

```
=====
PROC MONITOR - SYSTEM INFORMATION
=====
[1] CPU & MEMORY INFORMATION
-----
CPU Model : Intel(R) Core(TM) i7-12700H
Logical CPUs : 4
Total Memory : 3.30 GB
Available Memory : 1.09 GB
Free Memory : 0.25 GB
Memory Usage : 67.0%
-----
[2] CPU UTILIZATION & SYSTEM STATISTICS
-----
CPU Usage : 1.3%
System Uptime : 0 Days 0 Hours 32 Minutes
Load Average
  1 min : 0.07
  5 min : 0.17
 15 min : 0.12
-----
[3] RUNNING PROCESSES
-----
PID PPID STATE THREADS VmSize(MB) NAME
1 0 S 1 26 systemd
2 0 S 1 0 kthreadd
...
4093 2582 S 49 1484850 code
4419 4093 S 23 1484572 code
5332 4026 R 1 7 proc_monitor
-----
[4] TOP 5 MEMORY CONSUMERS
-----
Rank PID Memory(MB) Process
1 4419 357 code
2 4175 304 code
3 3983 267 ptyxis
4 2894 247 gnome-shell
5 4334 189 code
-----
TOP 5 CPU CONSUMERS
-----
Rank PID CPU(%) Process
1 2894 3.0 gnome-shell
2 3983 3.0 ptyxis
3 1188 2.0 dbus-daemon
4 1202 1.0 polkitd
5 4249 1.0 code
=====
```